

A Retrieval-First Pipeline for Single-Image Global Geolocalization, and the Autonomous Ratchet That Found It

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Abstract

We describe a system that predicts the geographic coordinates of a single street-level photograph with a median great-circle error of 34.1 km on a frozen 2,998-image validation set, and that runs entirely offline on a consumer Android phone. The system has two stages. A DINOv3 ViT-L/16 backbone adapted with rank-16 LoRA classifies the image over 5,000 learned geocells; the resulting posterior gates a nearest-neighbour search over an index of 1,198,072 geotagged training images; a learned listwise reranker reorders the top 100 candidates, and the winner’s coordinates are returned verbatim. The classifier alone reaches 104.7 km. Everything below that number was bought for a few dollars of compute in total, most of it spent on cached descriptors rather than on images.

Three findings drive the result. First, a geocell classifier is bounded by its own quantization: with 5,000 cells and an argmax readout the median cannot fall below 34.2 km, and retrieval removes that bound by returning a real coordinate rather than a cell centre. Second, the residual error is neither a data problem nor a gating problem: a training image within 25 km exists for 100% of validation queries and survives the gate 92 to 98% of the time, yet the matcher selects one only 46% of the time, so the entire remaining gap is matching quality. Third, cheap post-hoc operations on cached descriptors (hubness correction, whitening, multi-band contrastive heads and a sixteen-feature logistic reranker) are collectively worth more than any of the encoder fine-tunes we trained, and cost three orders of magnitude less.

The pipeline was found by an autonomous single-agent research loop that mutates one experiment file, runs it once against a frozen evaluator, and keeps the change only if the measured objective improves. We report the full ledger of 74 logged runs, including 35 rejected ideas and 9 crashes, and we treat the negative results as a primary contribution.

1 Introduction

Planet-scale image geolocalization asks a deceptively simple question: given one photograph and nothing else, where on Earth was it taken? No metadata, no panorama, no user context. The problem has been attacked in two broadly separate traditions. The retrieval tradition, opened by IM2GPS [18] and revisited in the deep learning era [37], matches the query against a large geotagged database and transfers the coordinates of the best match. The classification tradition, established by PlaNet [38] and refined by hierarchical and scene-aware variants [26, 10], partitions the globe into cells and trains a classifier over them. Recent work has largely converged on the classification branch: PIGEON [16] builds semantically-aligned geocells and applies haversine label smoothing, OpenStreetView-5M [3] studies the classify-then-regress hybrid at scale, and GeoCLIP [6] aligns image features to a continuous location encoder.

This paper argues, from measurement rather than from principle, that at the accuracy levels that matter for a practical product the classification branch is a means and the retrieval branch

is the end. A classifier is a good instrument for deciding *where to look*. It is a poor instrument for deciding *exactly where*, because its output is quantized by construction. Our classifier, trained on all 1.2 M images of a Street View corpus for three epochs, plateaus at a median error of 104.7 km. The same frozen features, reused as a retrieval index and searched under a posterior-mass gate, reach 78.5 km on the first attempt, and cheap post-processing on cached descriptors takes them to 42.7 km without ever touching an image again. Two short LoRA fine-tunes of the encoder and a seventeen-parameter reranker carry the full pipeline to 34.1 km.

The work has a second, methodological dimension that we do not think can be separated from the first. Every experiment reported here was proposed, implemented, launched and scored by an autonomous agent operating a keep-or-reset ratchet in the style of [23]: one experiment artifact is mutated, run once against a frozen evaluator, and kept only if the objective improved. The human’s role was to set the objective, to veto on grounds the metric could not see, such as model size for a mobile target, and to sign off on spend. That regime produced a specific and unusual research record: a complete ledger in which rejections are as carefully documented as acceptances, because the agent needed them to avoid re-proposing dead ideas. We report that ledger.

Contributions.

1. A two-stage pipeline reaching a 34.1 km median, 46.3% within 25 km and 10.1% within 1 km on a single static image with no metadata, scored by an evaluator that was written before the study began and never modified during it.
2. A decomposition of the residual error into coverage (100%), gate survival (92 to 98%) and matcher selection (46%), which identifies matching quality as the sole remaining bottleneck and rules out three otherwise plausible directions.
3. A set of cheap techniques that run on cached vectors (band-wise geo-smooth contrastive heads, candidate-side hubness correction, PCA whitening, and a sixteen-feature listwise logistic reranker) and that take the frozen-feature index from 65.7 to 42.7 km, and the final pipeline from 41.0 to 34.1 km, at approximately zero marginal GPU cost.
4. An honest negative-results section covering thirty-five rejected ideas, including several that improve every intermediate diagnostic while leaving the objective unchanged or worse.
5. An on-device realization: the entire encoder and a 300k-place index run offline on an Android phone at 33.9 km median, with a quantization study showing that int8 is unusable for this backbone and why.

2 Related work

Classification over geocells. PlaNet [38] framed geolocalization as classification over an adaptive partition of the globe and established the template that most subsequent work follows. Hierarchical variants [26] add coarse levels to catch wrong-continent errors, and [10] conditions on scene type. PIGEON [16] contributes two ideas we adopt directly: geocells constructed to respect administrative and semantic boundaries rather than pure density, and haversine label smoothing, in which the target distribution over cells is $\text{softmax}(-d/\tau)$ with d the great-circle distance from the true point to each cell centroid. We use $\tau = 75$ km throughout, and an independent sweep over $\{40, 75, 110\}$ km confirmed that value as the optimum for our cell count. PIGEON also reports that a retrieval refinement stage is responsible for most of their sub-kilometre accuracy, which is consistent with what we observe at much larger effect size.

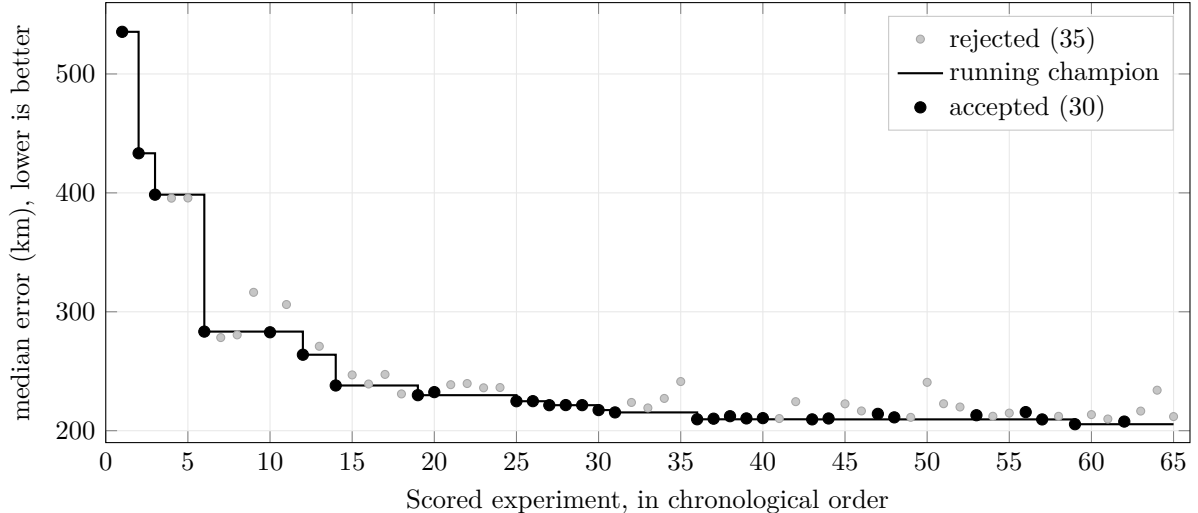


Figure 1: The autonomous ratchet on the proxy objective (60,000 training images, an eight-minute per-experiment budget, full 2,998-image validation split for scoring), plotted directly from the committed run ledger. All 65 scored runs are shown; the 9 crashes are omitted because they produce no number. The step function is the running champion, 535.4 $\rightarrow$ 205.5 km. The accepted set includes re-runs of the champion used to re-anchor after a change of rented machine, which is why some accepted points sit above the champion line. A few rejected points sit below it: a gain smaller than the acceptance bar of the moment was discarded as noise or held for a confirmation run, and the champion steps on the confirmation. The named changes behind each step are listed in Table 1. The three largest single drops are a throughput change, a cell-count change and a change to the prediction rule, in that order, and none of them is a change to the model architecture.

Classify-then-regress. OpenStreetView-5M [3] evaluates a broad design space and reports that predicting a cell and then regressing a within-cell offset outperforms either alone. We tried this twice, once as a per-cell tangent-plane regression head and once as a shared offset head, and it failed both times at our training budget. The reason is instructive and we return to it in Section 8: at a cell top-1 accuracy of 18%, the error is dominated by choosing the wrong cell, not by position within the right one.

Visual place recognition. Once the problem is reframed as retrieval, the relevant literature is visual place recognition. NetVLAD [2] and GeM pooling [29] define the aggregation baseline; CosPlace [4] and EigenPlaces [5] define the large-scale training recipes; SALAD [20] and BoQ [1] are the current descriptor state of the art. Patch-NetVLAD [17] and its successors perform geometric verification of the top candidates. Our failure mode, a Finnish forest road whose nearest neighbour is a near-identical Finnish forest road 300 km away, is perceptual aliasing, which is a place recognition problem rather than a geolocalization problem. That reframing was the most productive single conceptual move of the project.

Retrieval post-processing. We use cross-domain similarity local scaling (CSLS) [11] to correct hubness [30], and PCA whitening [21] on the raw descriptor space. We evaluated and rejected k -reciprocal re-ranking [39], graph diffusion, and SuperGlobal-style query expansion [31]. Learned reranking over a candidate set follows the Reranking Transformers line [36]; our reranker is a much smaller model fit on a few thousand queries.

Backbones and adaptation. The encoder is DINOv3 ViT-L/16 [32], the successor to DINOv2 [28], itself a Vision Transformer [13] with rotary position embeddings [33]. We adapt it with LoRA [19] at rank 16 and never modify the pretrained weights. PatchDropout [24] is used

during training only. Optimization is AdamW [25].

Concurrent work. Three recent papers surfaced during our own literature sweeps and influenced design choices without being reimplemented in full: a semivariogram-based formulation of geographic hard negative mining [7], HierLoc [14], which embeds a hierarchy of geographic entities in hyperbolic space, and Pinpoint [9], which reports a large gain from reranking on top of retrieval. The last of these was the direct motivation for the reranker described in Section 6.5.

3 Task, data and evaluation protocol

3.1 Task

One RGB image in, one (lat, lon) pair out. No panorama, no compass heading, no capture time, no camera metadata, no text prompt. This is a deliberately restrictive contract chosen because it is the contract a phone application can honour: a user supplies a photograph and nothing else.

3.2 Data

We use `josefbednar/streetview-acw-300k`, a corpus of Street View panoramas sampled world-wide. The training split contains 1,198,072 images from 299,518 distinct locations at four headings each; the validation split alone spans 123 countries. The held-out validation split contains 2,998 images, each drawn from a distinct panorama (verified by panorama identifier), so no validation image shares a location with another validation image, and none shares a location with the training split. Sampling is pinned to seed 1337 and cached once, so every experiment in the program sees exactly the same images.

For the fast experimental phase we use a frozen subset of 60,000 training images (approximately 15,000 locations at four headings), and a fixed 1,000-image validation subset for in-training monitoring. Every reported score, including in that phase, uses the full 2,998-image split.

3.3 Metrics

The primary objective is the median great-circle error in kilometres, computed with the haversine formula at $R = 6371.0088$ km. The median is chosen over the mean because the error distribution has a heavy tail: roughly 2.5% of images are sent to the wrong continent, and those alone move the mean by hundreds of kilometres. We report the mean as a secondary diagnostic and treat it, correctly, as a proxy for country-level accuracy rather than for fine-grained precision.

Alongside the median we report $\text{acc}@R$, the fraction of images within R kilometres of the truth, at $R \in \{1, 25, 200, 750, 2500\}$ km, corresponding roughly to street, city, region, country and continent resolution. We also report the GeoGuessr world score, $5000 \cdot e^{-d/1492.7}$, because it is the metric a user of the deployed application actually experiences. A rule of thumb worth keeping in mind throughout: a proportion near 46% measured on 2,998 images carries a 95% binomial interval of roughly ± 1.8 percentage points, so differences smaller than that between nearby configurations should be read as suggestive rather than settled.

3.4 The frozen harness

Scoring lives in a separate module that was written before the study began and never modified during it. It owns the data subset, the splits, the haversine implementation and the metric

panel. The model side exposes only a function mapping images to coordinates. The experiment file was the only artifact allowed to change during the search, and this was enforced structurally rather than by discipline: the launcher uploads only that one file to the compute node, so the node keeps the original evaluator regardless of what the agent edited locally. A score can therefore be improved only by making the model better, never by making the test easier.

The harness also owns the training deadline. It arms a wall-clock alarm before the training loop and raises an exception into the main thread at the budget, so every run terminates and reaches its final evaluation even if it hangs. This detail matters more than it looks: it is what makes an unattended overnight run produce a complete ledger rather than a partial one.

4 Methodology: an autonomous keep-or-reset ratchet

4.1 The loop

The search procedure is a ratchet. An agent reads the accumulated research log, selects one idea, edits the experiment file, runs it once on a rented GPU, parses the objective line from the run output, and accepts the change only if it beat the current champion by more than the noise floor. Rejected changes are reverted and recorded with the measured number and a short diagnosis. There is exactly one experiment in flight at any time. There is no population, no bandit, no surrogate model, and no parallel exploration. The only state carried forward is the champion file and a written log.

The simplicity is the point. It makes the record auditable: every number in this paper corresponds to one run of one file against one evaluator, and the file at that commit is in version control.

4.2 The proxy objective and its noise floor

The full training corpus is too large to iterate on. The experimental phase therefore ran on a proxy: 60,000 training images and an eight-minute training budget per experiment, scored on the full validation split. Two properties of this proxy were measured rather than assumed.

Run-to-run variation on the proxy is ± 3 to 5 km on the median. We initially required a gain of at least 8 km before crowning a change, and later relaxed the rule to 1 km with no confirmation re-runs, on the human’s instruction, after the acceptance rate collapsed. The relaxation was a deliberate trade of precision for throughput and it is visible in the record: several late-phase acceptances are within noise of each other.

More interesting, and less expected, is that the objective is not portable across rented machines. The identical champion file, run on rented instances in different data centres, scored 209.9 km and 214.2 km. Nothing about the model changed. We attribute this to differences in achieved step count within the fixed wall-clock budget, and it forces a protocol: after any machine change, re-run the champion to establish a local anchor, and compare only against that anchor. Sessions that skipped this step produced comparisons we later had to discard.

4.3 What the ratchet found

The proxy objective fell from 535.4 to 205.5 km, a reduction of 62%, across 74 logged runs of which 30 were accepted (a figure that includes re-runs of the champion used as baselines and machine anchors), 35 were rejected and 9 crashed. Figure 1 shows the trajectory and Table 1 names every accepted change in order.

The three largest single gains are worth naming, because none of them is an architectural idea. Removing gradient checkpointing and raising the worker count bought 19%, because at an eight-minute budget throughput is very nearly the objective. Growing the cell count from

Accepted change	Median (km)	Anchor (km)
Baseline: frozen DINOv3 ViT-L, LoRA $r=16$, 448 px, 512 geocells	535.4	
Remove gradient checkpointing, raise data-loader workers	433.3	
2048 geocells, average the top 16	398.5	
Mode-seeking readout: temperature 0.5, 1000 km locality	283.4	
Hierarchical heads at 64, 512 and 2048 cells	263.9	
Input resolution 448 to 384 px	238.0	
Fewer in-training evaluations	229.9	
Compile the backbone, warm up before the clock starts	224.8	
Larger real batch, made affordable by compilation	221.5	
Drop half the patch tokens in training forwards	215.4	
Double the batch again, spending the freed memory	209.6	
Country-level hierarchy with distance-smoothed targets	209.5	
Keep 60% of patch tokens rather than 50%	213.0	214.2
Median-seeking loss	209.5	215.7
Second, staggered tessellation	205.5	215.7

Table 1: Every accepted change, in the order it was accepted, with the proxy objective after it. The last three rows were measured on different rented machines, where the champion itself re-scores as shown in the anchor column; those rows are improvements against their own anchor, not continuations of the column above them.

512 to 2048 bought 8%, because the median was floored by cell size. Changing the *prediction rule* bought 29%, and is discussed in Section 5.3.

The eleventh row is the most instructive of the set and Figure 2 shows why. Dropping half the patch tokens in training doubled the number of optimizer steps that fit in the budget, and the objective improved. Training at a lower resolution and evaluating at full resolution raised the step count by 60% again, to 4410, and the objective got *worse*. Doubling the batch instead cut the step count back to 1500 and produced the best result of the whole phase. Step count is the currency only until roughly 2750 steps; past that point the currency is per-step quality.

4.4 What the ratchet did not transfer

We then trained the champion recipe on the full corpus for three epochs on four RTX 5090s, in two hours, and obtained a median of 104.66 km (Figure 3). A second full run, incorporating every subsequent accepted change, produced 104.37 km. Within the noise of a two-hour run, that is a wash.

This is the central negative result of the experimental phase, and it deserves to be stated plainly. Improvements found under a fixed eight-minute budget on a 5% data subset do not reliably compound at three epochs on the full corpus. Several of them, the throughput unbrake in particular, are answers to a question that only exists inside the proxy. Two substantially different recipes landing on the same 104 km strongly suggests that the binding constraint at scale was the training regime itself, epochs and adapter capacity, rather than the loss or the tessellation. That conclusion is what redirected the entire project to inference-time methods, and every result after this point is a consequence of that redirection.

5 Stage one: the geocell classifier

5.1 Architecture

The encoder is DINOv3 ViT-L/16 at 384×384 , giving 576 patch tokens plus five prefix tokens with rotary positions. The pretrained weights are frozen. Adaptation is LoRA at rank 16 on the q , k , v and o projections of all 24 blocks, 3.15 M parameters. A trunk of LayerNorm, a

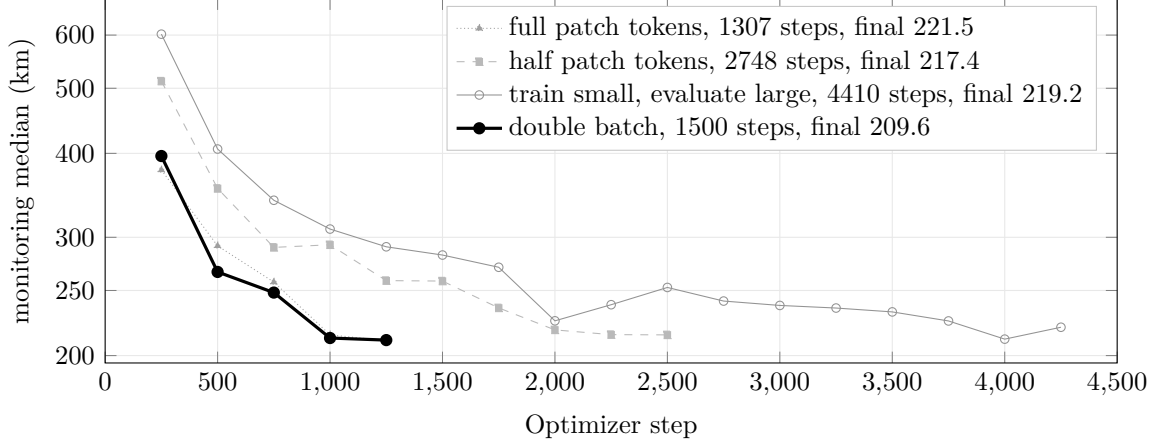


Figure 2: Step count stops being the currency. Curves are the in-training monitoring evaluation on the fixed 1,000-image subset, logged live during four experiments that each had the identical eight-minute budget; the number quoted in each legend entry is that run’s final score on the full validation split, which is the metric decisions were made on. The run with the most steps by a wide margin is not the best, and the run with the fewest is. The half-token run shown here is the first of its pair; its confirmation re-run scored 215.4 km, which is the figure that enters Table 1.

1024 $\rightarrow$ 1024 linear layer and GELU sits on the CLS token.

Above the trunk are four heads. A fine head classifies over 5,000 geocells obtained by country-constrained k -means over training coordinates. A second fine head classifies over an independent 5,000-cell tessellation from a different seed, and the two are unioned at readout so that the Voronoi boundaries of one do not align with those of the other. Coarse heads at 64 and 512 cells are added into the fine logits through child-to-parent maps. A country head over 115 countries is added the same way, through each cell’s majority country; its count is smaller than the 123 countries the data spans because a country enters this head only by being the majority in at least one cell. Total trainable parameters are 15.2 M, approximately 5% of the backbone’s weight count.

5.2 Losses

The fine and coarse heads are trained with cross-entropy against distance-smoothed targets, $p^*(c) \propto \exp(-d(y, \mu_c)/\tau)$ with $\tau = 75$ km and μ_c the cell centroid, following [16]. The country head uses the same construction over country centroids with $\tau = 300$ km, which measurably outperformed one-hot country targets: near-miss countries should not be penalized as heavily as antipodal ones.

To this we add a median-seeking loss. Let $w = \text{softmax}(\ell/T)$ over the fine logits ℓ with $T = 0.5$, and let

$$\hat{v} = \frac{\sum_c w_c u_c}{\|\sum_c w_c u_c\|_2}, \quad (1)$$

where $u_c \in \mathbb{S}^2$ is the unit vector of cell centroid c . This is a differentiable spherical mean, the sphere-valued analogue of soft-argmax [34]. We penalize its great-circle distance d to the truth with a redescending Geman–McClure robust loss [15],

$$\mathcal{L}_{\text{med}} = \frac{d^2}{d^2 + s^2}, \quad s = 300 \text{ km}, \quad (2)$$

weighted at 0.5 against the cross-entropy terms. Two properties motivate this. It closes the mismatch between a training objective on cells and a test-time readout that is a spherical mean over cells. And its gradient vanishes for hopeless cases, so the loss optimizes something much

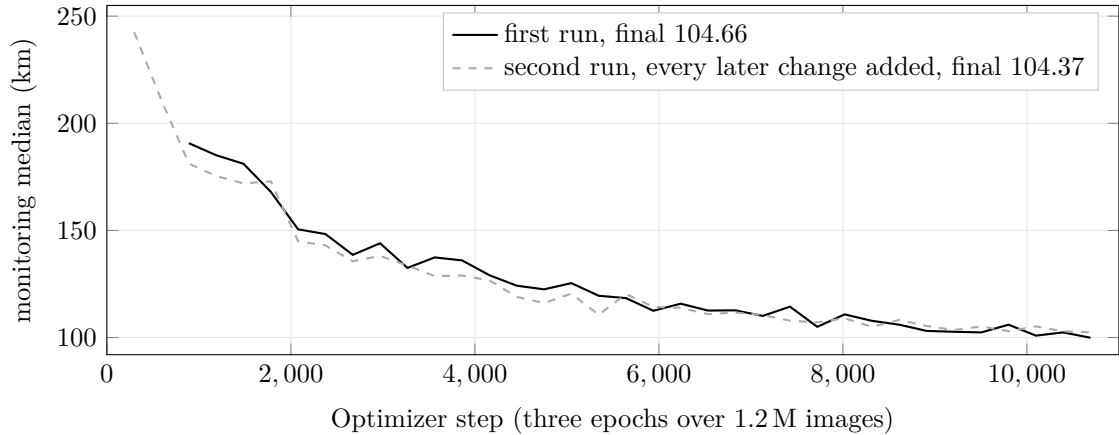


Figure 3: Two full-corpus training runs, logged live on the 1,000-image monitoring subset; the finals quoted in the legend are on the full validation split. The second run incorporates every change accepted after the first was launched, and lands in the same place. Both curves are still descending at the three-epoch cutoff, which says the binding constraint is the length of the run rather than the recipe.

closer to the actual median than cross-entropy does. Empirically it improved the objective by 6.2 km on the proxy while leaving cell top-1 accuracy exactly unchanged, which is precisely the signature of a pure mass-placement improvement.

5.3 The readout, and why the median rewards mode seeking

The single largest gain of the entire experimental phase, 29%, came from changing how a posterior is converted into a coordinate. The natural readout, a probability-weighted spherical mean over all cells, is wrong for a multimodal posterior. If a model places 40% of its mass in Finland and 35% in Canada, their spherical mean is in the North Atlantic, which is worse than either. The fix is to sharpen with temperature $T = 0.5$ and restrict the average to the top- k cells lying within 1000 km of the top-1 cell.

The mean error worsens slightly under this rule and the median improves enormously, which is exactly as expected. It is worth stating as a general lesson: under a median objective, a prediction rule should seek modes, not averages, and this is a property of the metric rather than of the model.

5.4 The quantization floor

A classifier with a hard argmax readout can never do better than its own tessellation. On our 5,000 country-constrained cells the measured floor is 34.2 km: a hypothetical perfect classifier, always selecting the correct cell and always emitting its centroid, would score exactly that. Our classifier reaches 104.7 km with a cell top-1 accuracy of 27.4%.

Two corollaries followed and both were tested. First, the soft spherical-mean readout can go below the floor, since it interpolates between centroids, so 100% top-1 accuracy is not even optimal under our prediction rule. Second, finer cells raise the ceiling but make the classification problem harder: moving to 4,096 cells in the proxy phase improved acc@25 and worsened the median, because a 4,096-way cross-entropy does not converge in the available steps. We swept the cell count and closed the question.

The productive conclusion is that removing the floor requires leaving the classifier’s output space entirely. That is what the second stage does.

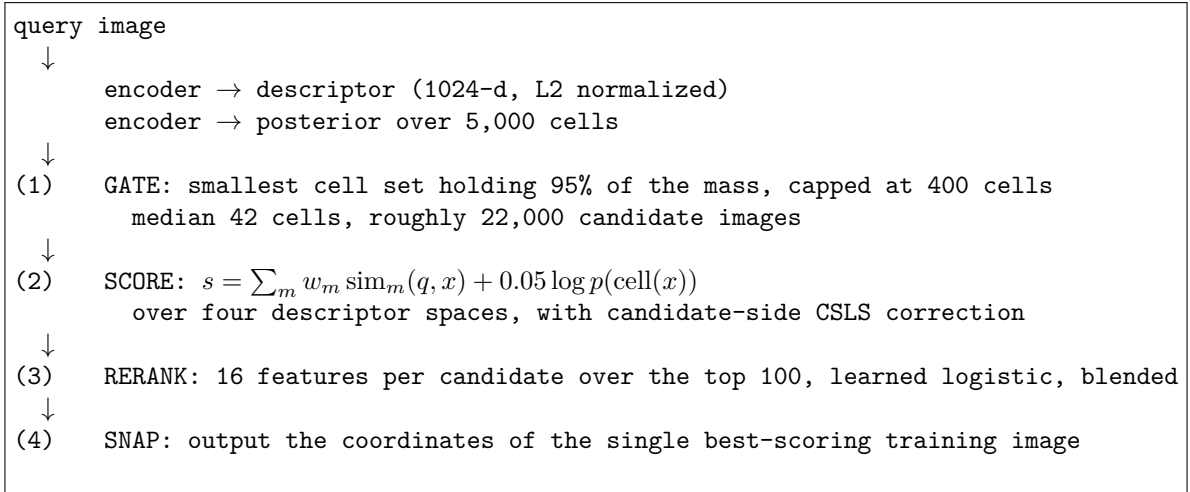


Figure 4: The inference pipeline. Stage one supplies both the descriptor and the gate. Only step (4) produces a coordinate, and it is always a real training coordinate, never a cell centre.

6 Stage two: gated retrieval

Figure 4 shows the shape of the second stage: the classifier’s posterior selects a small set of cells, a blended similarity scores the images inside them, a learned model reorders the best hundred, and the winner’s coordinates are the answer.

6.1 Gating by posterior mass

The index holds one descriptor per training image, 1,198,072 vectors, stored in fp16 and bucketed by cell. Searching all of it for every query is both slow and harmful: distant distractors that happen to look similar are the dominant error mode.

Rather than a fixed top- K of cells, we gate by posterior mass. We take the smallest set of cells whose posterior sums to 0.95, capped at 400 cells. The resulting set has a median size of 42 cells and contains the true location within 100 km for 91.6% of queries, which is far better than any fixed K at comparable pool size. The mass threshold turns out to matter very little: 0.90, 0.95 and 0.99 score identically to three decimal places, which is itself informative, since it says the posterior’s shape rather than its calibration is doing the work.

The classifier is then given a weak vote inside the pool through a log-prior term with weight $\lambda = 0.05$. A sweep found $\lambda = 0$ and $\lambda = 0.15$ both worse. The weight is small on purpose: a confident but wrong country prediction must be overruling by a strong visual match.

Finally, the readout is a hard snap, $k = 1$. We tested similarity-weighted averaging over the top- k neighbours, consensus voting, location max-pooling and cluster centroids. Every one of them was neutral or worse. Trusting the single nearest match wins, which is again a property of the median.

6.2 Which features to index

An early and load-bearing finding: the descriptor should be tapped from the *backbone CLS token*, before the geo-classification trunk, not after it. The post-trunk representation is worse in every top configuration. The interpretation is that training a trunk to classify continents actively discards the instance-level detail that distinguishes one Finnish road from the next, which is exactly the detail retrieval needs. The two branches of the network want different things from the same encoder, and only the shared backbone should be common.

A second finding contradicted our expectations completely. We re-embedded the entire index at 512 px, at $1.8\times$ the compute, expecting a meaningful gain. It was worth +0.33 percentage points of acc@25 . Resolution is not the lever, and the classifier side corroborates this: 384 px beat 448 px there three separate times.

6.3 Place heads and the geo-smooth correction

Raw DINOv3 cosine similarity answers the question “do these two photographs look alike”, which is the wrong question. Two photographs of the same thirty metres of forest road taken 180° apart look nothing alike, while a Finnish road and a Canadian road look identical.

We therefore learn a projection. Each place head is a small residual MLP, $1024 \rightarrow 2048 \rightarrow 1024$, with the output layer zero-initialized so training starts from the identity, fit contrastively with NT-Xent [8, 27] on *cached* descriptors. No images are touched and no backbone gradients are computed, so a head costs roughly one minute of GPU time and about fifteen cents.

The first head used same-location positives and improved held-out view-recall from 84.9% to 99.0%, but improved the end metric only modestly. The diagnosis, and the correction it implies, is the most transferable idea in this paper. At test time the query’s own location is never in the index. The best reachable answer is therefore a *different* location one to fifteen kilometres away, and a same-location objective explicitly pushes those neighbours apart. Retraining with positives drawn from any location within d_{pos} kilometres, and masking in-batch negatives closer than 50 km as false negatives, moved the median from 61.2 to 49.0 km.

The choice of d_{pos} is a bias-variance trade rather than a hyperparameter to be tuned once. We train a *band* of heads at $d_{\text{pos}} \in \{5, 10, 25\}$ km and blend their similarities with the raw space at weights (0.34, 0.22, 0.22, 0.22). A single head trained at an intermediate radius does not reproduce the blend: we tried it, and it lost. Diversity across bands, not the average band, is what the blend exploits.

6.4 Hubness correction and whitening

High-dimensional nearest-neighbour search suffers from hubness [30]: a small number of index items are the nearest neighbour of a disproportionate share of queries. In a geolocalization index those hubs are aliasing attractors, generic scenes that pull queries from everywhere.

We apply cross-domain similarity local scaling [11] on the candidate side. For each index item x we precompute $r(x)$, the mean cosine similarity to its ten nearest neighbours among 5,000 pseudo-queries sampled from the index, and score

$$s_{\text{CSLS}}(q, x) = 2\text{sim}(q, x) - r(x). \quad (3)$$

This took roughly forty seconds of arithmetic on cached vectors and was, until the reranker of Section 6.5 arrived, the largest of the cheap gains: 43.75 $\rightarrow$ 39.94 km, with acc@25 rising from 44.20 to 44.90%.

Independently, we PCA-whiten the raw CLS space [21], fitting the mean and covariance on a 200,000-vector sample of the index and renormalizing afterwards. Whitening alone gives 41.00 km. The three levers, band blending, CSLS and whitening, are close to orthogonal and stack to 36.96 km.

6.5 Learned listwise reranking

The final component is a learned reranker over the gated pool, and it is the largest inference-time gain in the pipeline. For each query we take the top $K = 100$ candidates by blended score and build sixteen features per candidate.

Let s_k be the blended score, bb_k the raw CLS cosine, and $w = \text{softmax}(s/0.05)$ over the pool. Let d_{jk} be the great-circle distance between candidates j and k , and C_{jk} their CLS cosine affinity. The features are: s_k ; bb_k ; $\log(1 + \text{rank}_k)$; the margin $s_1 - s_k$; three *geographic consensus* terms $\sum_j w_j \mathbf{1}[d_{jk} \leq R]$ for $R \in \{10, 25, 50\}$ km; a one-step diffusion of w over the candidate affinity graph thresholded at $C > 0.5$ and row-normalized; the affinity-weighted mean $\sum_j w_j C_{jk}$; the mean of the three largest off-diagonal affinities of candidate k ; the entropy of w ; the pool fill fraction; and four interaction terms pairing s_k , bb_k , the margin and the diffusion term with the 25 km consensus feature. Features are standardized over the flattened candidate set.

The model is a logistic regression fit by full-batch gradient descent, with binary target $\mathbf{1}[d(x_k, y) \leq 25 \text{ km}]$, evaluated under five-fold cross-validation over queries so that no query is ever scored by a model that saw it. The final ranking score is

$$\tilde{s}_k = \frac{s_k}{\text{std}(s)} + \lambda f(\mathbf{x}_k), \quad (4)$$

with f the fitted logit and λ a blending weight.

Two things about this are worth dwelling on. The first is how small it is. Sixteen features and one logistic regression, fit on 2,998 queries, moved the median from 40.98 to 34.10 km, a 17% reduction, for zero GPU cost. We had previously tried an MLP reranker at the same data scale and it failed: capacity without data. The second is what the winning features encode. The three consensus terms measure whether a candidate is surrounded by other plausible candidates in *geographic* space. A cluster of mutually-close candidates beats a lone high-similarity outlier. That is a hypothesis about the structure of the failure mode, namely that aliasing produces isolated confident matches while true matches produce geographically coherent neighbourhoods, and it is directly testable, which is why we tested it as a standalone feature first (-4 km on its own) before building the full model.

The blending weight λ behaves as an interpretable dial between the two accuracy regimes. At $\lambda = 1$ the system scores 36.66 km median with 11.61% within 1 km; at $\lambda = 2$ it scores 34.10 km with 10.14% within 1 km. Since the target is binary at 25 km, this is expected: pushing harder on the 25 km objective costs sub-kilometre precision. A graded target $\propto \exp(-d/\tau)$ should remove the trade and is the obvious next step.

6.6 Encoder fine-tuning

Three attempts were made to improve the encoder itself, on the reasoning that all of the above can only re-weight information the frozen backbone already retains.

The contrastive encoder. The first attempt fine-tunes the backbone with LoRA, warm-started from the trained classifier, on a purely contrastive objective: positives within 10 km, negatives beyond 50 km, in-batch pairs between the two masked out, batches restricted to a single ~ 500 km region so that negatives are visually similar but geographically wrong, and embeddings gathered across all four GPUs for a large negative pool. Trained for 2,500 steps, or 0.53 epochs, in 42 minutes. The raw descriptor’s acc@25 rose from 36.32 to 42.46%, the largest single-component jump of the project.

It also produced a rule we applied to everything that followed. A place head trained *online*, inside the fine-tune, scored 33.89%, ten points worse than the same recipe fit *offline* on cached descriptors afterwards. Train encoders online; fit heads offline.

Two things went wrong. The classifier degraded, because the contrastive objective rode on the shared trunk: gating with the fine-tuned model’s own logits scored 42.93% against 44.20% using the previous checkpoint’s logits, so inference temporarily needed two encoder forward passes. And the binary loss, which treats 80 km away exactly like another continent, improved sub-kilometre accuracy while degrading the 25 to 200 km band.

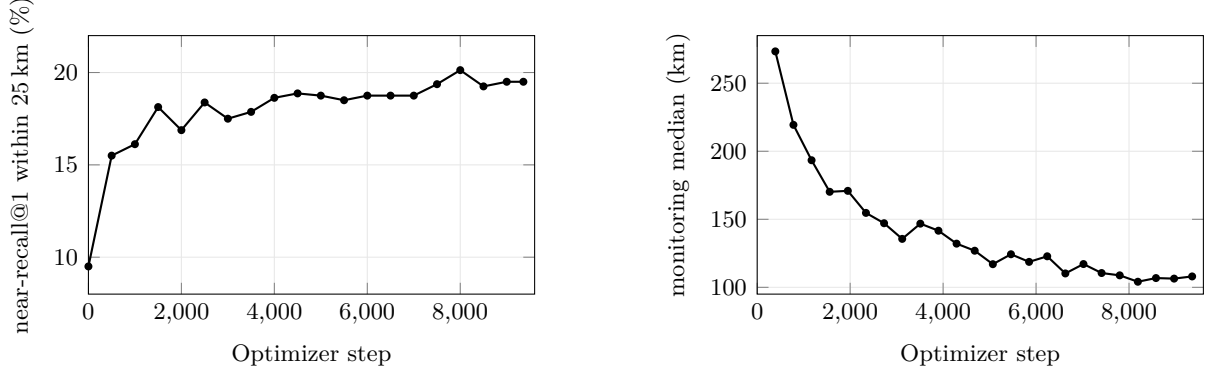


Figure 5: The from-scratch joint encoder, both objectives logged live over its single epoch. Left: the held-out retrieval probe, which doubles from 9.5% to 19.5%. Right: the classifier’s monitoring median, which reaches 104.1 km, matching a three-epoch classifier run in one joint epoch from scratch. Both curves say the recipe works. Its descriptors nevertheless lose to the two warm-started encoders, which is the point of Section 7.3.

The jointly trained encoder. The second attempt fixes both. The binary contrastive loss is replaced by a graded one, using soft targets $\propto \exp(-d_{ij}/\tau_{\text{geo}})$ with $\tau_{\text{geo}} = 100$ km, which is the same smoothing the classifier already uses and removes the artificial equivalence between near-misses and antipodes. The classifier cross-entropy is trained jointly as an anchor, batches mix region-restricted and global samples, and LoRA is extended to the MLP blocks. The joint objective worked as designed: this model’s classifier is measurably better than the one from the second full-corpus run (103.19 against 104.47 km median, cell top-1 27.12 against 26.35%),¹ and a single forward pass now serves both the gate and the descriptor.

The from-scratch encoder. The third attempt rebuilds everything in one joint phase with no checkpoint lineage, so that every number is attributable to a single recipe. It adds attribute heads supervised by labels derived for free from the coordinates (driving side, climate class, hemisphere, latitude band) and a graded listwise loss over candidate pools mined from geography alone, with climate-matched cross-continent negatives aimed at the confidently-wrong tail. Figure 5 shows it working exactly as intended on both of its objectives.

The honest outcome of all three is reported in Table 4 and discussed in Section 7.3: none of them beat the cheap post-processing, and the final champion’s advantage is attributable to the reranker rather than to any encoder.

7 Results

7.1 Main result

The final system scores a median great-circle error of 34.10 km, with 46.26% of images placed within 25 km and 10.14% within 1 km, at a mean of 328 km and a GeoGuessr score of 4470 per round. Table 2 places this in context.

We include PIGEON’s published numbers because it is the closest comparable system and because they were the target we set ourselves, but they are measured on a different evaluation set, and we make no claim of a controlled comparison. The internally controlled comparison is the one against the same author’s prior system, which used the identical 1.2 M-image corpus

¹Figure 3 quotes that run at 104.37 km. The numbers here come from re-scoring both checkpoints side by side under the retrieval phase’s evaluation code, which lands a tenth of a kilometre away from the training run’s own final readout.

System	median (km)	@1 km	@25 km	@200 km	mean (km)
<i>Reported on other evaluation sets, not directly comparable</i>					
PIGEON, single image [16]	131	0.9%	24.2%	–	–
PIGEON, four-view panorama [16]	44.4	5.4%	40.4%	–	–
<i>This work, all on the same frozen 2,998-image split</i>					
Prior system, full fine-tune, same corpus	194	–	–	–	–
Classifier stage only	104.66	0.0%	7.6%	70.6%	456
Full pipeline, $\lambda=1$	36.66	11.61%	45.86%	–	–
Full pipeline, $\lambda=2$	34.10	10.14%	46.26%	–	328
On device, int8 index, 1 view per location	44.4	–	43.0%	73.9%	–
On device, int8 index, 4 views per location	33.9	–	46.4%	74.6%	–

Table 2: Headline results. The PIGEON rows are included for orientation only; they are measured on a different held-out set and a different image distribution, and no claim of a controlled comparison is made. The on-device rows are measured with the exact quantized assets the application ships, on the same validation split.

Stage	median (km)	@1 km	@25 km	mean (km)
(a) Classifier, full corpus, 3 epochs	104.66	0.0%	7.6%	456
(b) + retrieval: mass gate, cosine kNN, snap	78.52	10.24%	35.16%	–
(c) + tuned gate, prior $\lambda=0.05$, $k=1$	65.66	10.11%	35.99%	–
(d) + contrastive place head, cached vectors	61.16	11.47%	38.69%	390
(e) + geo-smooth positives, two-head blend	49.04	11.57%	41.16%	388
(f) + mean-patch descriptor, chamfer rerank	43.75	11.27%	42.93%	373
(g) + contrastive encoder, bands, CSLS, whitening	36.96	12.78%	46.06%	378
(h) + joint encoder, learned reranker ($\lambda=2$)	34.10	10.14%	46.26%	328

Table 3: The full progression on the frozen validation split. Rows (b) through (h) are inference-time only with respect to the classifier: it has not been retrained since row (a). Rows (b) to (g) cost approximately \$6 of rented GPU time in aggregate; row (h)’s reranker cost nothing beyond CPU minutes. The chain is cumulative with one exception: the configuration in row (h) retires the chamfer rerank introduced in row (f), whose benefit no longer justified its cost once the learned reranker was in place.

with a full backbone fine-tune and scored 194 km. Against that, with 5% of the trainable parameters, the classifier stage alone is 46% better and the full pipeline is 82% better.

7.2 Stage-by-stage ablation

Table 3 gives the complete progression. The shape of it is the argument of this paper: the largest single jump, $104.66 \rightarrow 78.52$ km, comes from reusing already computed features as a retrieval index, with no training at all.

7.3 Encoder attribution

Table 4 is the table we would most like other people to copy, because it makes an uncomfortable attribution explicit. Our champion configuration uses the jointly trained encoder, but that encoder’s descriptors are *worse* than the purely contrastive one’s under an identical recipe, 40.98 against 36.96 km. It became the champion because it is the index the reranker happened to be fit and evaluated on, and the reranker is worth more than the difference between the two encoders. The correct reading is that the 34.10 km result is the reranker’s win, not the encoder’s.

One cell of the table is missing, and it is the one we most regret. The purely contrastive index, the best of the four under the matched recipe, never received the reranker: its cached candidate pools were lost with the machine that produced them, before the reranker existed.

Encoder behind the index	recipe only		+ learned reranker	
	median (km)	@25 km	median (km)	@25 km
None: frozen classifier features	49.04	41.16%	42.74	43.1%
Contrastive, warm-started	36.96	46.06%	not measured	—
Joint with graded targets, warm-started	40.98	44.13%	34.10	46.26%
Joint, from scratch, one epoch	47.82	42.56%	38.65	45.10%

Table 4: The four indices compared at a matched post-processing recipe (band blend, hubness correction, whitening), and again with the learned reranker on top. The reranker figures use $\lambda = 1$ for the frozen-feature index and $\lambda = 2$ for the other two. The champion configuration is the jointly trained encoder plus the reranker, even though that encoder’s descriptors are worse than the purely contrastive one’s at matched recipe.

Regenerating them costs a few dollars, and doing so is the first measurement anyone extending this work should take, because it could plausibly beat 34.10.

The from-scratch encoder is a cleaner negative, and the more interesting one, because it succeeded at everything it was measured on during training and still lost. In a single joint epoch its held-out retrieval probe doubled, from 9.5% to 19.5% near-recall within 25 km, and its classifier reached 104.1 km, matching a three-epoch classifier run (Figure 5). Its descriptors nevertheless lose to both warm-started encoders, which carry roughly three and a half epochs of accumulated backbone adaptation between them. A corroborating diagnostic: its descriptor space has a mean hubness statistic of 0.787 against the joint encoder’s 0.604, that is, it is measurably under-differentiated, with more of the index acting as a nearest neighbour to everything. The conclusion is narrow but useful: one joint epoch from scratch is enough to match the classifier and not enough to match the descriptors, so the two objectives converge at different rates on the same backbone.

7.4 Where the remaining error lives

Three ceilings, all measured on the same validation split, decompose the residual error and close off three directions.

Coverage is 100%. For every single validation image there exists a training image within 25 km, and the median nearest training image is 2.2 km away, with 38.5% of queries having a training image within 1 km. More data cannot raise this ceiling, though denser coverage would still make the matcher’s task easier. This was measured before any retrieval work began and is what justified the entire second stage.

In-pool oracle is 92 to 98%. Restricted to the candidates the gate actually shows, a perfect matcher would place 92% of queries within 25 km at a mass threshold of 0.8 and 97.6% at 0.95. The gate is not the constraint either.

The matcher converts 46%. That is the number we actually achieve.

So approximately fifty percentage points of acc@25 sit entirely in matching quality. Not in coverage, not in the gate, not in the cell scheme. Two full rounds of head engineering moved it from 38.7 to 42.9% with sharply diminishing returns, which is the evidence that the limit lies in the features themselves. A backbone tuned to tell Finland from Canada keeps what distinguishes Finland from Canada and discards what distinguishes one Finnish road from the next.

There is a further arithmetic consequence worth stating, because it sets the target for whoever continues this. The moment acc@25 crosses 50%, the median is mathematically below 25 km. One lever moves both metrics.

within	share	within	share	percentile	error
100 m	2.97%	50 km	51.83%	p10	0.8 km
500 m	7.74%	200 km	73.92%	p25	5.4 km
1 km	11.27%	1000 km	94.50%	p50	43.8 km
5 km	24.15%			p75	211 km
10 km	31.72%			p90	564 km
25 km	42.93%			p99	7986 km

Table 5: Full distance profile at the multi-descriptor stage, median 43.75 km, on all 2,998 validation images. We report the profile at this stage because it is the configuration for which the complete per-image distance array was archived. The GeoGuessr score at this configuration is 4438 per round, 4856 median round, 22,188 per five-round game, with 69.3% of rounds above 4500.

System	@1 km	@25 km	@200 km	@750 km	@2500 km	training imagery
ISNs [26]	10.5	28.0	36.6	49.7	66.0	4.7M Flickr
GeoCLIP [6]	14.1	34.5	50.7	69.7	83.8	4.7M Flickr
PIGEOTTO [16]	11.3	36.7	53.8	72.4	85.3	4.5M Flickr+landmarks
Ours, classifier argmax	0.0	3.2	17.4	38.0	60.7	1.2M Street View
Ours, full pipeline	0.5	9.6	20.3	41.2	65.0	1.2M Street View
Oracle: nearest index image	13.6	86.8	90.6	98.3	100.0	(index coverage bound)

Table 6: IM2GPS3k, percentage of queries localized within each radius. Published rows are from the cited papers; all are trained on Flickr-distribution corpora, the same distribution the benchmark is drawn from. The oracle row is the fraction of queries for which *any* index image lies within the radius: the ceiling of any retrieval over our index.

7.5 Distance profile

Table 5 gives the shape of the error distribution. The distribution is strikingly bimodal in practical terms. A quarter of all queries land within 5.4 km, which is usable-street-level accuracy, and one query in a hundred lands nearly eight thousand kilometres away. There is very little in between, and this is why the mean is a poor summary and the median a good one.

7.6 External benchmarks: transfer to consumer photographs

Every number so far is measured on held-out Street View imagery. To measure transfer to consumer photographs directly, we ran the frozen champion pipeline on the two standard benchmarks of the field: IM2GPS [18] (237 Flickr images) and IM2GPS3k [37] (2,997 Flickr images). The protocol involves no benchmark-side fitting of any kind: the encoder, gate, blend weights, hubness correction, whitening and the E7 reranker (fit once on our own validation split) are all applied exactly as deployed.

Table 6 reports IM2GPS3k; IM2GPS is qualitatively identical (median 660 km, 10.1% at 25 km). The result is poor, and the decomposition says precisely why. Index coverage is not the bottleneck at city scale: an index image within 25 km exists for 86.8% of IM2GPS3k queries. But the matcher converts that coverage 11% of the time, against 46% in-domain. A Street View-trained descriptor is being asked to match a Flickr close-up, an indoor scene or a landmark photograph against road-level views of the same place, and mostly cannot. At street scale the index itself caps the task: only 13.6% of queries have *any* index image within 1 km, so even a perfect matcher could not reach GeoCLIP’s 14.1%. Two facts survive the transfer intact. Retrieval still triples the classifier (3.2% → 9.6% at 25 km, median 1498 → 1144 km), so the structural claim of this paper, classify to look and retrieve to answer, holds out of domain. And the failure is a corpus property rather than a method property: the published systems

above train on the benchmark’s own distribution, while the two Street View-trained systems in the literature simply do not report these benchmarks (PIGEON proper was never evaluated on them, and OSV-5M [3] declines the comparison, judging roughly a third of IM2GPS3k non-localizable). We report the numbers anyway: they are the honest measurement of the domain gap that Section 10.3 previously only asserted.

8 Negative results

The ledger contains thirty-five rejected ideas. We group the informative ones.

Auxiliary classification heads improve every diagnostic and not the objective. Geo-contrastive hard negatives, a per-cell prototype cosine head, a surgical rerank in which a fused score selects cells while the original posterior weights the spherical mean, and an exponential moving average of the trainables were each tried. Every one of them raised cell top-1 and top-5 accuracy, in one case to the best values recorded in the whole program, and every one left the median flat or worse. The lesson generalizes: cell top-1 is a pessimistic proxy for a median objective, because a one-point gain in top-1 moves too few validation rows to shift the fiftieth percentile. We stopped proposing ideas in this family after the fourth independent confirmation.

Within-cell refinement fails while cell selection is bad. A per-cell tangent-plane offset regression head, evaluated twice under different parameterizations, and a barycentric soft-label scheme designed to be exactly invertible by the readout, both lost. At a cell top-1 of 18%, regressing position inside a 250 km cell is about as hard as the classification itself, and refining within a cell that is usually the wrong cell cannot help. This directly contradicts the OSV-5M [3] finding at their scale, and we believe the reconciliation is simply that classify-then-regress requires a converged selector.

Query-side augmentation is dead in retrieval. Test-time augmentation with three centre crops cost 9.3 km on the classifier. Query-side flips and zooms cost 0.4 percentage points of acc@25. A multi-resolution ensemble combining the 512 px and 384 px indices of the *same* model cost 0.9 points, because the same model’s errors at two resolutions are correlated and there is no diversity to exploit. Database-side augmentation remains untested and is the asymmetric version we would try next.

Batch samplers that reduce location diversity are catastrophic. A panorama InfoNCE auxiliary using same-panorama pair batches cost 23 km. A distillation scheme grouping four views per location cost 30 km. An online confusion-driven sampler upweighting confused country pairs cost 7 km. Three independent confirmations of the same law: at a fixed step budget, distinct locations per batch is the scarce resource, and any sampler that spends it loses more than the auxiliary signal returns.

Classical retrieval post-processing does not transfer. We implemented k -reciprocal re-ranking [39] in two fair variants, visual and geographic graph diffusion, and SuperGlobal-style query expansion [31] in two variants. All were neutral or negative. Our reading is that these methods assume a database in which true matches form dense mutual neighbourhoods, whereas a worldwide Street View index is dominated by generic scenes for which that assumption fails.

No post-hoc decision rule can fix the mean. This is the most decisive negative result in the project and it took six experiments to establish. The mean is owned by roughly 5% of

queries where both evidence channels, the descriptor space and the classifier posterior, point confidently at the wrong region. We tried falling back to the classifier (twice, the second time with corrected centroids), jumping to the training-density prior, shrinking toward the prior, snapping to a geometric median, and re-gating at mass 0.995 for low-confidence queries. All failed, and they failed for a reason that generalizes: the outcome-selected tail is not the detector-selected tail. Queries the system is uncertain about are ones it already handles decently, whereas the truly catastrophic misses are *confidently* wrong, so no confidence threshold isolates them. Progress on the mean requires new evidence channels, not new arithmetic on the existing ones.

Miscellaneous closed questions. The smoothing temperature τ was swept at 40, 75 and 110 km and closed at 75. The cell count was swept at 512, 2048, 4096 and 5000, and at the experimental budget closed at 2048. Learning rates were swept and closed. LoRA rank 32 lost to rank 16, because capacity costs steps. A hexagonal balanced tessellation classified far better, top-1 25.7% against 18%, and guessed far worse, +26.5 km, which is the quantization floor made visible. Muon [22] on the head matrices at learning rate 0.02 lost by 26 km, which we read as a hyperparameter failure rather than a statement about the optimizer.

9 Deployment

The pipeline runs entirely offline on an Android phone. Nothing about the deployment is a demonstration harness: it is the same encoder, the same gate, the same blend, the same CSLS correction and the same reranker, ported to Kotlin and verified against the Python reference on identical inputs, with 32 of 32 test images producing an identical top-1 match at a maximum divergence of 0.000 km.

9.1 Accuracy and cost of quantization

Two index configurations ship. A one-view-per-location index of 1.55 GB scores 44.4 km median and 43.0% within 25 km. A four-view index of 6.18 GB scores 33.9 km and 46.4%. The four-view figure matches the research pipeline’s own 34.10 km, which means int8 compression of the *index* costs nothing measurable: the entire gap between the two configurations is the three missing views per location, not the quantization.

9.2 Massive activations make int8 unusable for this encoder

Quantizing the *encoder* is a completely different story, and the finding is worth recording because it is not specific to this application. DINOv3 ViT-L carries a massive-activation channel in its residual stream, peaking around 157,000, of the kind documented in large language models [35, 12]. Per-tensor dynamic int8 quantization collapses the cosine similarity between the on-device descriptor and the index descriptor from 0.989 to 0.596, and in the whitened space to 0.12. The same outlier overflows fp16 and yields NaN under a naive half-precision export.

The export that works stores all 156 linear layers in fp16 while keeping the LayerNorms, the residual path and the patch-embedding convolution in fp32. This is numerically identical to fp32 at half the size, 652 MB against 1.3 GB, and is what ships.

9.3 Latency

Table 7 makes the engineering point plainly: searching the one-view index costs 156 ms and encoding one image costs 3.7 s. The four-view index multiplies the search term by four at most, which changes nothing about the balance. Retrieval, the component that produces almost all of the accuracy, is essentially free at this scale, and the entire latency budget belongs to the

	Pixel 10 Pro XL	Galaxy S21+	S21+ thermally throttled
Encode	3.7 s	6.4 s	14.9 s
Search, one-view index	156 ms	470 ms	750 ms
Startup	1.1 s	1.7 s	7.8 s

Table 7: On-device latency, with search measured on the one-view index of 299,518 vectors. Retrieval is not the bottleneck; the ViT-L forward pass is, and it scales directly with sustained CPU clock.

backbone. The throttled column is instructive: sustained inference heat-soaks an Exynos 2100 from 2912 to 533 MHz, and every number degrades by that same factor, which is a useful reminder that on-device benchmarks without a thermal column are not benchmarks.

One implementation detail cost a day and is worth passing on. The `createScaledBitmap` call on Android is bilinear, while the index was built with Pillow’s bicubic resampler. The resulting descriptor drift is small in pixel terms and large in retrieval terms. The shipped preprocessing reimplements Pillow’s bicubic resampler by hand.

10 Discussion

10.1 Why retrieval wins here

The result admits a compact explanation. A classifier over N cells has an output space of size N ; its predictions are, up to the interpolation the soft readout provides, confined to N points on a sphere. A retrieval system’s output space is the set of training coordinates, which here has 299,518 distinct elements covering every validation query to within 2.2 km on the median. The second output space is simply much better matched to a kilometre-scale objective, and no amount of classifier improvement changes that.

What the classifier is genuinely good at is deciding where to look. Its posterior, gated at 95% mass, reduces 1.2 M candidates to about 22,000 while retaining a within-25 km candidate 92 to 98% of the time. That is an excellent instrument, and it is exactly the instrument that a pure retrieval system lacks.

10.2 The economics of cached-vector research

A theme that runs through the whole project is that experiments on cached vectors cost cents while experiments involving images cost dollars, and the ratio is roughly three orders of magnitude. The entire progression from 104.7 to 36.96 km cost approximately \$9 of rented GPU time. The reranker that took it to 34.10 km cost nothing but CPU minutes on a laptop.

This suggests an operating discipline that we came to late and would adopt from the start next time: the product of a run on rented hardware is not a model, it is a cache. The complete set of artifacts, descriptors for both splits, logits, the gated candidate list per query, candidate pools for reranker training, attribute posteriors, is what allows every subsequent experiment to run on a CPU in minutes. The last of our encoder runs was designed explicitly around that contract, with a manifest checked before the machine was allowed to be destroyed. The encoder it produced lost. The cache it produced is still being used.

10.3 Limitations

The headline evaluation is on a single held-out split of one Street View corpus, and Street View imagery has a distinctive and consistent capture process. Section 7.6 now measures transfer to consumer photographs directly, and the measurement is sobering: 9.6% at 25 km on IM2GPS3k

against 34–37% for Flickr-trained systems. The decomposition there shows the gap is dominated by descriptor domain shift rather than index coverage, which bounds what any amount of post-hoc lever-pulling on the current corpus can recover. Closing it requires consumer-distribution imagery in both the fine-tune and the index; it is a data problem before it is a method problem.

The reranker is fit and evaluated by cross-validation on the same 2,998 queries. The folds are disjoint by query and no query is scored by a model that saw it, so the numbers are not optimistically biased in the usual sense, but the feature set and the blending weight λ were selected with knowledge of this split. We would not defend the third significant figure of 34.10.

The mean, at 328 km, is not good, and Section 8 explains why we believe no rearrangement of the current evidence will improve it. The system fails confidently, which is the worst failure mode for a user-facing product, and a genuine confidence estimate remains unsolved.

Finally, our comparison to published systems is uncontrolled. Different evaluation sets, different image distributions. We have been careful to label it as orientation rather than as a result.

10.4 Privacy

A system that turns an arbitrary photograph into coordinates is useful and abusable in equal measure: the capability that lets a hiker identify a trailhead also lets a stranger narrow down a home from a shared snapshot. The authors of PIGEON [16] discuss this tension at length and chose not to release their strongest model, and we take their reasoning seriously. For privacy purposes the honest summary of our system is not the median but the tail of Table 5: roughly one photograph in ten resolves to within a kilometre of where it was taken, and that is the number a release decision should weigh. Two further facts bear on such a decision. The deployed pipeline runs entirely offline, so no safeguard can be added on a server after the fact. And the index derives from Street View imagery, so any redistribution is bounded by the licensing terms of the source imagery rather than by ours.

10.5 Notes on autonomous experimentation

Three observations from running a research program this way.

The frozen evaluator is not a formality. It is the only thing that makes an autonomous agent’s self-reported results trustworthy, and enforcing it structurally, by uploading only the mutable artifact to the compute node, is much stronger than enforcing it by instruction.

The negative record is the expensive artifact. An agent that does not remember its rejections re-proposes them, and the accumulated dead-end list was consulted more often than the champion configuration. We would now design the log format before designing the loop.

Instrument the training loop with the metric the final decision uses. Our contrastive fine-tune ran blind for 42 minutes: its contrastive loss swings by a factor of two purely on the geographic difficulty of the sampled region and therefore forecasts nothing, and its five intermediate checkpoints could not be ranked without a full re-index each. A two-second held-out retrieval probe would have provided both a forecast and a checkpoint selection criterion. We added one before every subsequent run.

10.6 Future work

The ceiling analysis in Section 7.4 points at exactly one target: within-pool matching quality, currently 46% against an oracle of 92 to 98%. Three directions follow directly. Replacing CLS pooling with an optimal-transport aggregation over patch tokens [20] is the largest literature-endorsed descriptor gain available and we already compute and discard the necessary patch tokens. Geometric verification of the top ten candidates [17] attacks perceptual aliasing directly,

since true neighbours share scene geometry while 300 km lookalikes share only statistics. And the reranker should be scaled: the version reported here is a sixteen-feature logistic fit on three thousand queries, and a train-side episode cache of 1.2 M mined pools already exists for fitting a much larger one with a graded rather than binary target.

For the mean, the direction is new evidence channels rather than new decision rules: attribute probes over free coordinate-derived labels used as inference-time constraints on the gate, and a script-identification or OCR channel, which is high-precision where it fires.

11 Conclusion

We set out to place a single photograph on the globe and reached a median error of 34.1 km, with the whole system running offline on a phone. The result was obtained by treating a geocell classifier as a search-space reducer rather than as a predictor, and by spending almost all of the remaining effort on cheap operations over cached vectors: contrastive projections fit offline, hubness correction, whitening, and a small learned reranker whose most valuable features measure geographic consensus among candidates rather than visual similarity to the query.

The measurement we would most like to be remembered is the ceiling decomposition. Coverage is perfect, the gate is nearly perfect, and the matcher converts less than half of what it is handed. Everything that matters in this problem, at this accuracy level, is in the matcher.

Acknowledgements

The experimental loop is a fork of `karpthy/autoresearch` [23], whose keep-or-reset design and frozen-harness discipline are the reason the ledger in this paper is trustworthy. All experiments were proposed, implemented and executed by an autonomous agent under human supervision; the author set the objective, approved spending, and vetoed on grounds the metric could not see.

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A Experiment ledger summary

The complete ledger is committed as a tab-separated file with one row per run, recording the commit hash, the objective, peak memory, the verdict and a one-line diagnosis. Its summary statistics are given in Table 8.

Outcome	Runs
Accepted (including baselines, machine anchors and confirmation re-runs)	30
Rejected	35
Crashed (out of memory, distributed deadlocks, scripted-edit bugs)	9
Total logged	74

Table 8: Outcomes across the whole search phase. Crashes are logged because their causes, memory pairing rules and collective-communication timeouts, are reusable knowledge.

Nine crashes is not a small number and we report it deliberately. Six were out-of-memory failures, which established a standing rule that any memory-adding change must be paired with a batch-size reduction in the same experiment. Two were distributed-training deadlocks caused by running a diagnostic probe on all ranks, desynchronizing them past the collective timeout. One was a scripted-edit bug that closed a class scope and silently removed a method, after which every scripted edit to the model file was checked by parsing the resulting syntax tree.

B Reranker feature list

For completeness, the sixteen per-candidate features of Section 6.5, in order: blended gated score; raw CLS cosine; $\log(1 + \text{rank})$; margin to the top candidate; geographic consensus at 10, 25 and 50 km; one-step affinity diffusion of the score softmax; affinity-weighted mean similarity to other candidates; mean of the three largest off-diagonal affinities; pool entropy; pool fill

Component	Trained by	Parameters
DINOv3 ViT-L/16 backbone	never	300 M (frozen)
LoRA adapters, $r=16$, on q, k, v, o	full run, then contrastive	3.15 M
Trunk, fine heads A/B, coarse and country heads	full run	12.0 M
Band place heads	offline, cached descriptors	12.6 M
Learned reranker	offline, cached candidate pools, CPU	17
Total trained		27.8 M (9%)

Table 9: The complete parameter budget. The reranker, which is worth 6.9 km of median error, is a sixteen-weight logistic regression with a bias term.

fraction; and the four interactions of score, raw cosine, margin and diffusion with the 25 km consensus term. Features are standardized across the flattened candidate set. The softmax temperature is 0.05, the affinity graph is thresholded at cosine 0.5 and row-normalized, and the logistic is fit by 400 steps of full-batch gradient descent at learning rate 0.5 under five-fold cross-validation over queries.

C Reproducibility

The classifier, the retrieval engine, the offline head-training scripts, the lever sweeps and the local evaluation scripts are all in the project repository, along with the research log, the raw run ledger and the working notebooks. Checkpoints and cached descriptor arrays are mirrored to a public model repository, folder per phase. The evaluator was never modified during the study; the commit that introduced it predates every result in this paper.

Every run logged its full metric panel live to an experiment tracker, one run per experiment. Figure 1 is plotted from the committed run ledger, and Figures 2, 3 and 5 are plotted from the tracker’s recorded histories rather than redrawn by hand, so the curves are the ones the decisions were actually made against. Where a figure shows the in-training monitoring evaluation, which uses a 1,000-image subset, the corresponding full-split score is quoted alongside it, because the two differ by a few kilometres and only the latter was ever used to accept or reject a change.

D Parameter budget

Table 9 accounts for every parameter the final configuration trains. We include it because the distribution is the argument of the paper in miniature: almost all of the accuracy comes from the last two rows, and the last row has seventeen parameters.